

AR(1) Bivariate Normal Distribution

$$\underline{X} = \begin{pmatrix} X_t \\ X_{t-1} \end{pmatrix}$$

$$\underline{X} \sim N_2 \left(\begin{pmatrix} E(X_t) \\ E(X_{t-1}) \end{pmatrix}, \text{Var}(\underline{X}) \right)$$

Variance-covariance matrix of $\underline{X}$

$$\text{Var}(\underline{X}) = \begin{pmatrix} \text{Var}(X_t) & \text{Cov}(X_t, X_{t-1}) \\ \text{Cov}(X_t, X_{t-1}) & \text{Var}(X_{t-1}) \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\sigma_z^2}{1 - \phi_1^2} & \frac{\sigma_z^2 \phi_1}{1 - \phi_1^2} \\ \frac{\sigma_z^2 \phi_1}{1 - \phi_1^2} & \frac{\sigma_z^2}{1 - \phi_1^2} \end{pmatrix}$$

$$\Gamma \text{ If } \begin{pmatrix} X \\ Y \end{pmatrix} \sim N_2 \left(\begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \begin{pmatrix} \sigma_x^2 & \rho \sigma_x \sigma_y \\ \rho \sigma_x \sigma_y & \sigma_y^2 \end{pmatrix} \right)$$

then

$$X|Y=y \sim N \left(\mu_x + \frac{\sigma_x}{\sigma_y} \rho (y - \mu_y), \sigma_x^2 (1 - \rho^2) \right)$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

Here

$$X_t | X_{t-1} = x_{t-1} \sim N \left(\mu + \phi_1 (x_{t-1} - \mu), \sigma_z^2 \right)$$